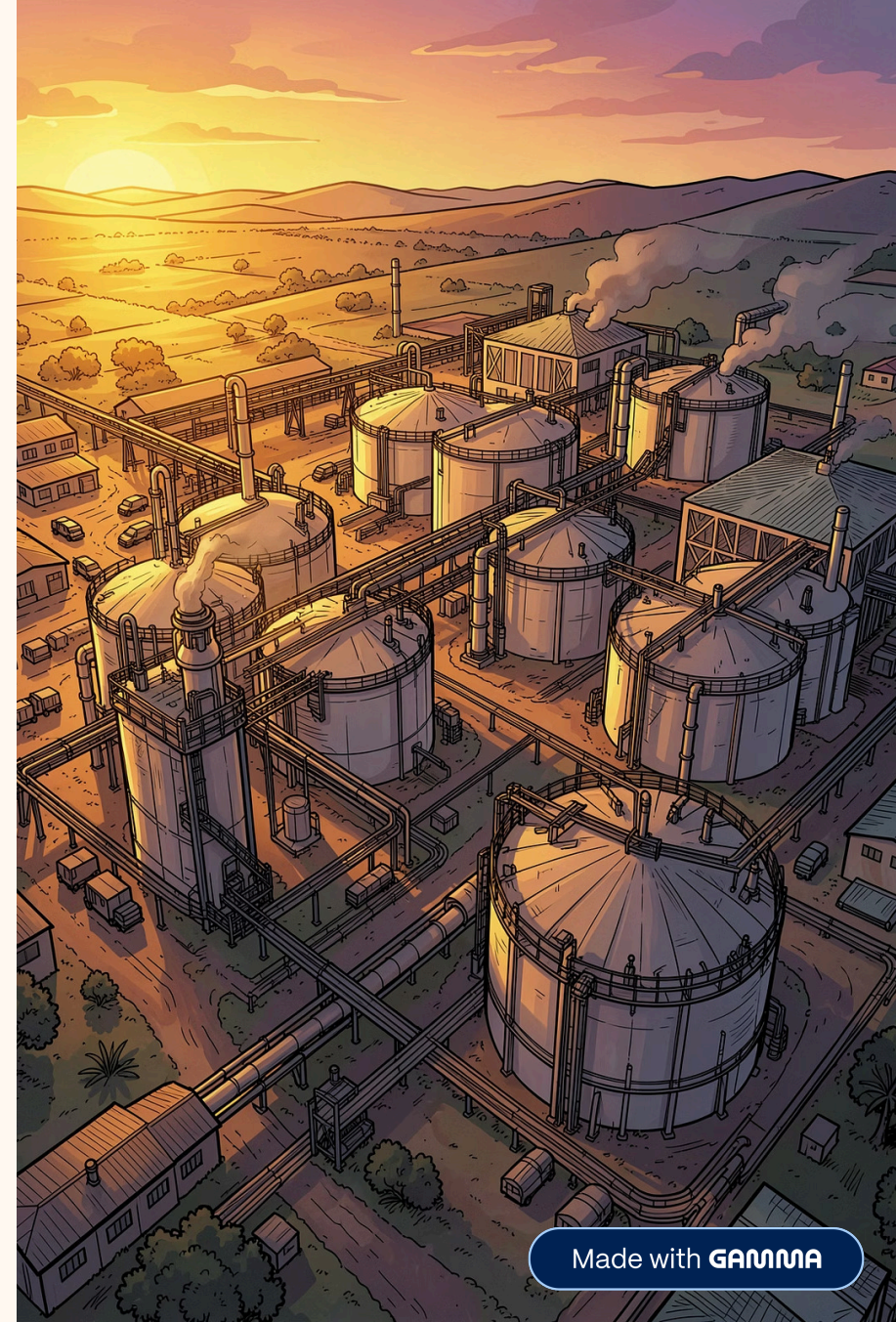


# The Operational Question

Managing fuel depots across Kenya means monitoring pipeline pressure as a critical operational metric. Two hypotheses drove this analysis:

- Does rainfall correlate with reduced throughput?
- Do public holidays affect pipeline pressure?

Answering these questions enables better forecasting, reduces unplanned maintenance costs, and improves supply chain planning.



# Three Sources, One Master Dataset

## Source Breakdown

### Internal CSV

4,000 operational records: pressure, flow rate, temperature, and incident logs

### Open-Meteo Weather API

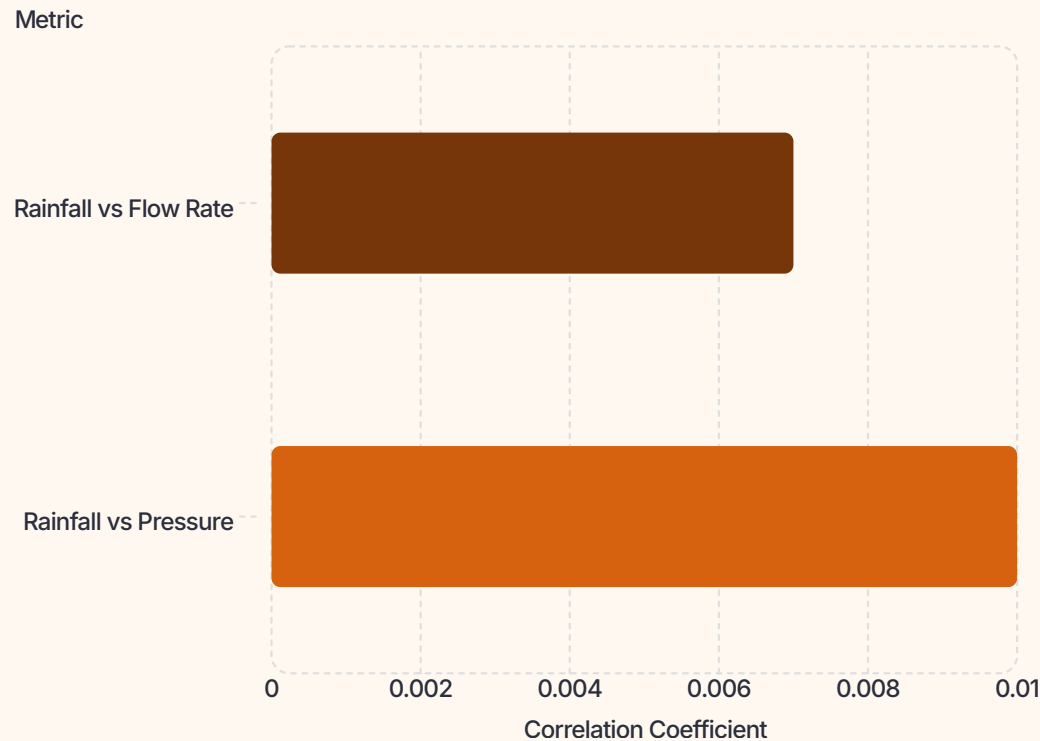
53 days of environmental data: temperature, precipitation, and wind speed

### SQLite Holidays Table

Public holiday records joined via GROUP BY for daily aggregates

All sources merged on `event_date` → **Master DataFrame: 4,000 rows × 22 columns**

# Weather Has No Significant Operational Impact



## Key Findings

### Rainfall → Flow Rate

Correlation: **0.007** —  
effectively zero

### Rainfall → Pressure

Correlation: **-0.010** -  
negligible

### Holiday Effect

Not testable ; no holidays fell within the 53-day window

## Recommendations

- **Redirect maintenance focus ;**  
weather is not a significant driver;  
prioritise mechanical and  
operational factors
- **Automate the pipeline ;** deploy  
continuous monitoring to detect  
anomalies in real time
- **Re-run with holiday data** extend  
the date range to capture public  
holidays for a complete analysis